
Absence Is Not Omission: Availability, Vintage, and the Measurement of Selective Benchmark Disclosure

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Abstract

1 Model providers choose which benchmark results appear in a release doc-
2 ument, while an independent evaluator scores the same models on many
3 more. Whether the left-out results stand below the printed ones seems
4 answerable with a bound on what could have been reported, and bench-
5 mark publication dates supply one: an instrument published after a model
6 shipped was not publicly available to it. The comparison this sets up fails
7 as a check on itself. Benchmarks that postdate a release already trail those
8 that predate it by 12.23 percentile points before any disclosure label enters,
9 because the date deciding whether a benchmark could have been reported
10 also fixes where a model sits inside that benchmark’s comparison pool, an
11 age-period-cohort collinearity that reweighting softens but cannot remove.
12 The same failure appears on a leaderboard we did not build: across fourteen
13 HELM Lite releases, a fixed set of models keeps identical published scores
14 while every win rate moves, and a companion leaderboard with a pool-
15 invariant headline shows none of it. Hand-coding every locatable release
16 document, checked by an independent second coding, makes the danger
17 concrete: the coded omission gap is +11.4 points, significant against zero,
18 yet it lands where the label-free gap and the share of reported benchmarks
19 predict. What survives is composition: providers print the benchmarks
20 they stand well on.

21 1 Introduction

22 When a frontier model ships, its provider publishes a model card carrying benchmark results,
23 and independent evaluators run standardised suites over the same models [Liang et al., 2023,
24 Epoch AI, 2026]. This paper asks whether benchmark availability can identify whether the
25 results a provider leaves out stand below the ones it prints, the older economic question.
26 Disclosure unravels only when the sender’s evidence is commonly known [Grossman, 1981,
27 Milgrom, 1981]. Dye [1985] shows what breaks that: a receiver who cannot tell a sender hold-
28 ing nothing from one holding something bad pools the two, and withholding survives. Any
29 study of what was left out inherits that pooling, since the object it needs is the endowment,
30 the results a sender held when it wrote, which the work below does not observe.

31 Benchmark documentation is one of the few settings where that endowment can be bounded
32 from outside: absent prepublication sharing, a benchmark published after a model shipped
33 cannot have been in its endowment, whatever a provider ran privately, so dating the in-
34 struments separates absence that could have been a disclosure from absence that could
35 not.

36 Our first contribution is a panel built around that bound. From a pinned snapshot of an
37 independent benchmarking hub covering 819 model-versions, we collapse scaffold variants
38 to the release a provider ships, giving 353 releases across 46 organisations, 74 benchmarks
39 and 3,082 scored pairs. Every benchmark carries a verified release date, splitting the panel
40 into 2,355 eligible pairs and 476 that could not have been disclosed.

41 Our second contribution is that bounding the endowment is not enough, since judging with-
42 holding needs a measure of how results stood against peers. The eventual comparison is
43 reported versus omitted benchmarks among those available, and it needs the two sides of
44 availability balanced in standing. They are not. Benchmarks that already existed when
45 a model shipped outscore benchmarks built afterwards by 12.23 percentile points on aver-
46 age, positive in 78.1% of the 146 releases holding both kinds of cell, and nothing in that
47 calculation knows which benchmarks a provider reported. The most favourable measure
48 of standing we can construct still leaves 9.14 points. The contamination is structural, be-
49 cause the date identifying the endowment also shifts the measure of standing used to price
50 silence [Heckman, 1979]. Read as economics: instrument vintages are a free verification
51 technology for the endowment that Dye [1985] shows a receiver cannot otherwise see; the
52 verifying variable is not excludable from the receiver’s valuation of silence, so what it verifies
53 it contaminates.

54 Our third contribution takes the question outside our own panel. Across 14 releases of HELM
55 Lite, the pool grows from 30 models to 91 while 24 models keep bit-identical scores, and
56 every one of their published headline values moves, five pairs reversing order. A companion
57 leaderboard from the same team, behind an absolute headline, shows none of this.

58 Our fourth contribution codes the record. Hand-coding the 100 releases with a locatable
59 release-time document, blind-replicated on a fifth at $\kappa = 0.95$, yields the estimator such a
60 study would use: omitted benchmarks stand +11.4 points above those a provider could not
61 have reported, excluding zero yet short of its matched label-free counterpart by only what
62 composition implies: it cannot distinguish concealment from the availability gap. What
63 survives is composition: reported benchmarks stand 8.7 points above omitted eligible ones
64 within release ($p = 0.0001$), a gap benchmark fixed effects do not remove.

65 1.1 Related literature

66 That reporting practice in model documentation is uneven, incomplete and shaped by the
67 sender is established, and we concede it [Stauffer et al., 2025, Kuehnert et al., 2026]. Three
68 papers sit closest. Zhang et al. [2026] examines four flagship releases inside a long-context
69 evaluation study and finds a benchmark family absent from every main result table, an
70 omission count without the dates that would tell a passed-over benchmark from one the
71 release predates. Ghosh et al. [2026] builds the reporting format in which an omission
72 would become visible, and asks what a reader could see rather than what was left out.
73 Singh et al. [2025] documents selection inside a live leaderboard, providers testing private
74 variants and keeping the best, alongside evaluator policies that move the ranking; what it
75 cannot separate, and we make separable at the cell level, is an omission from a result that
76 could not have existed. Bean et al. [2025] reviews 445 benchmarks for construct validity,
77 whereas we take the instruments as given and ask which results a provider prints. None of
78 the four conditions on availability, the single thing we add: without a verified benchmark
79 date, an omission and a nonexistence are the same observation. The economics runs from
80 unravelling under a known endowment [Grossman, 1981, Milgrom, 1981] to its failure under
81 endowment uncertainty [Dye, 1985] and under costly disclosure [Verrecchia, 1983], frictions
82 this design does not separate. The report-card literature supplies the cautionary cases
83 [Dranove and Jin, 2010, Hotz and Xiao, 2013, Jin and Leslie, 2003, Bederson et al., 2018].
84 Concurrent theory models strategic evaluation and persuasion directly [Truong et al., 2026,
85 Dai et al., 2026]; that work is theory, ours measurement.

86 2 Data and Setting

87 2.1 The disclosure problem

88 A release i ships at date t_i carrying a single document. Let $\mathcal{B}$ index benchmarks, each with
 89 its own publication date τ_b , let $E_i \subseteq \mathcal{B}$ be the endowment, the benchmarks whose result the
 90 provider holds when it writes, and let $D_i \subseteq E_i$ be the disclosed set, which reports a score
 91 s_{ib} for each $b \in D_i$. The receiver observes D_i and its scores and must split the complement
 92 into results held back and results never held.

93 Under a commonly known endowment, silence is priced at the worst value it could conceal
 94 and disclosure unravels to $D_i = E_i$ [Grossman, 1981, Milgrom, 1981]. Two distinct frictions
 95 break that. In Verrecchia [1983] the endowment is known and disclosure is costly, so with-
 96 holding is a cutoff in s_{ib} . In Section 3 of Dye [1985] disclosure is free but the receiver cannot
 97 tell an empty endowment from a concealed one, because a sender can prove possession and
 98 cannot prove emptiness.

99 Here an independent evaluator publishes s_{ib} for cells the provider passed over, and each τ_b
 100 makes infeasibility verifiable. Writing $A_i = \{b : \tau_b < t_i\}$ for the available set and O_i for the
 101 cells the evaluator scores,

$$D_i \subseteq E_i \subseteq A_i, \quad \theta = \Pr(b \notin D_i \mid b \in O_i \cap A_i).$$

102 A provider may hold results the evaluator never produced and a benchmark can exist and
 103 go unrun, so θ is defined against the evaluator’s coverage rather than the true endowment,
 104 and the Dye friction survives inside A_i .

105 Judging what was withheld then requires a measure of standing P_{ib} , since raw scores are not
 106 comparable across benchmarks. The design rests on an exclusion restriction: conditional on
 107 the innocent explanations, membership in A_i does not enter the equation for P_{ib} [Heckman,
 108 1979]. A date rule cannot satisfy it: $t_i - \tau_b$, t_i and τ_b are exactly collinear, so if any
 109 two of the three move P_{ib} the third cannot be excluded from it. All three move it here,
 110 which Section 3 shows and Appendix B bounds. Two things follow, and they are not the
 111 same. The first is practical: estimating θ needs disclosure labels, which the coded record
 112 of Section 5 supplies. The second is structural: reading θ as evidence of withholding runs
 113 into A_i itself: availability is a deterministic threshold in benchmark maturity, $\mathbf{1}[t_i - \tau_b > 0]$,
 114 so no cell of the same maturity sits on both sides of the boundary, and wherever standing
 115 depends on maturity, conditioning on maturity finely enough to make availability excludable
 116 destroys positivity. So availability alone cannot globally identify selective disclosure from
 117 any pool-relative standing measure, win rate or percentile alike, and a jointly estimated
 118 scale inherits the same selection through which cells exist. What survives, under continuity,
 119 is the discontinuity at the threshold.

120 2.2 The independent score set

121 The evidence base is a pinned snapshot of Epoch’s benchmarking hub taken on 17 August
 122 2026, covering 819 model-versions. The unit of disclosure is a release, keyed on organisation,
 123 model name, and release date. Epoch’s version field splits a release across scaffolds such as
 124 reasoning-effort and context-length variants, but a provider ships one document per release,
 125 so scaffold rows are collapsed to their release, keeping the best score across variants, with
 126 the spread kept as a diagnostic. The resulting panel holds 353 releases, 74 benchmarks and
 127 3,082 release-benchmark pairs for which an independent score exists. Providers are the unit
 128 of dependence throughout.

129 2.3 Eligibility and benchmark dates

130 A provider cannot omit a benchmark that did not yet exist. Every benchmark therefore
 131 carries its own release date, the earliest verifiable public release, taken from Epoch where it
 132 publishes one and hand-collected from the benchmark’s primary source otherwise, resolving
 133 preprint, repository and versioned dates to the earliest public appearance, and a pair is
 134 eligible only when the benchmark strictly predates the release and its date is verified. The
 135 bound is an assumption about access: an instrument shared before publication defeats it,

violations understate rather than manufacture the gap by moving available cells into the postdating group, and the coded record carries no trace of them, no document reporting a benchmark before its public release date. Eligibility yields 2,355 of the 3,082 pairs. Pairs whose benchmark postdates the release form a separate set of 476, the cells that could not have been disclosed. Pairs with no verified date are dropped rather than imputed, and error in the dates themselves attenuates the boundary estimate of Section 3. The release-level contrast imposes no minimum benchmark count, since requiring one would discard exactly the releases whose only cells are a single eligible and a single postdating one. Of the 146 releases carrying both kinds, 57 hold fewer than eight scored benchmarks. A second and simpler obstacle, that the number of benchmarks a release could have reported is itself set by when it shipped, is treated in Appendix A.

Our disclosure-coding protocol uses ORBIT, the Outcome Reporting Bias In Trials instrument from clinical trials, where a protocol-specified outcome vanishing from the published report has a long measurement literature [Chan et al., 2004, Kirkham et al., 2010, Dwan et al., 2013]. Present cells are graded by whether the document gives a number, names the benchmark without one, or substitutes a variant; absent cells by whether a same-family predecessor reported it, contemporaries did, or no comparable provider did either. Of a queue of 108 releases, the 103 with eligible cells in the pinned panel plus five from a later vintage, the 100 with a locatable release-time document are coded under it, 1,280 cells in all; no result before Section 5 rests on a disclosure label, and that section reports what the labels return.

2.4 Measuring standing

Raw scores are not comparable across benchmarks, so standing is measured as a within-benchmark percentile computed against contemporaneous models, meaning those whose release dates fall within 182 days on either side of the focal release, with midranks for ties. The peer count and the share of the window released after the focal model are retained alongside the percentile. This is the measure a windowed comparison naturally suggests, and the one the paper goes on to show is biased in a way that mimics selective disclosure.

2.5 Inference

Inference is constrained by the same structure. The panel has 46 nominal provider clusters, but 24 contribute fewer than ten cells, 20 appear in a single release, and the five largest hold 68.1% of the sample. Cluster-robust asymptotics are not credible below twelve clusters, and we report no analytic standard error there [Cameron et al., 2008]. Regression coefficients in that range use the restricted wild cluster bootstrap over 9,999 draws, which imposes the null before resampling Rademacher weights at the provider level. Release-level means use a cluster sign-flip randomization test over 9,999 draws, reversing an organisation’s entire contribution at once; it preserves provider dependence but not benchmark dependence, which the two-way clustered regressions carry, and their intervals are percentile bootstraps over 10,000 provider resamples. Both report $(k + 1)/(\text{draws} + 1)$, with k the draws at least as extreme as the observed statistic. Mirrored sign vectors leave 2^{G-1} distinct statistics over G clusters, so a resolution of 0.0001 needs fifteen clusters, and the 18 organisations behind the headline estimate clear it; in the regression range below twelve clusters the bootstrap switches to a six-point weight distribution whose finer support restores the resolution [MacKinnon and Webb, 2016, 2018].

3 The Availability Gap

3.1 The placebo comparison

Separating results held back from results never held calls for a control group whose endowment is known to be empty [Dye, 1985]. The dates appear to supply one: a benchmark not yet published when the model shipped could not, absent prepublication access, have been reported, while an eligible one could have been. Whether it was reported is what a disclosure label would say, and none enters here. Whatever makes an eligible benchmark look bad for

innocent reasons, difficulty drift, saturation, or a shift in what the field measures, presses on both sets alike, so under those explanations the difference should be zero. One innocent channel is asymmetric by construction: a benchmark published before a model shipped can sit in its training corpus and a later one cannot, so contamination raises eligible standing only, and no reweighting of the comparison can excise it.

It is not zero. Across 146 releases with both kinds of cell, eligible benchmarks outscore postdating ones by 12.23 percentile points on average, with a median of 10.69, and the gap is positive in 78.1% of releases. Nothing in that calculation knows which benchmarks were reported. A measure that balances the two sides of the window, described below, cuts the figure to 9.14 points, positive in 70.9% of 141 releases, without removing it. A cluster sign-flip randomization test over the 18 organisations holding both kinds of cell puts that mean at $p = 0.0001$, the smallest value the draws can return. Cells that could not have been reported already stand lower than cells that could, so the group is not a control.

3.2 Decomposing the gap

Table 1 locates the failure, regressing standing on a placebo indicator, equal to one when the benchmark postdates the release, always within release. With release effects alone, the placebo coefficient is -13.529 (2.038). Adding benchmark fixed effects absorbs 63.4% of it and leaves -4.949 (2.183). Benchmark composition therefore carries most of the gap, and structurally so: the panel is unbalanced, and placebo cells concentrate by construction in benchmarks that entered late. Those benchmarks are less saturated: split at median publication date, 8.3% of scores on the later half of the 54 benchmarks with ten or more scores sit in the top decile of the benchmark’s own range against 13.2% on the earlier half ($p = 0.015$), so a late instrument leaves more room below the ceiling. Adding the peer-window asymmetry term, the share of a cell’s comparison window released after the focal model, instead of benchmark effects absorbs 39.8% and leaves -8.151 (1.865). Conditioning jointly on benchmark effects and both window terms absorbs all but 0.003 points and leaves -0.003 (1.748), an interval reaching $+3.4$, so the residual is bounded at under a third of the original gap rather than at zero.

Conditioning set	Placebo coefficient	t	Absorbed
Release fixed effects only	-13.529 (2.038)	-6.637	0.0%
Release FE + benchmark fixed effects	-4.949 (2.183)	-2.267	63.4%
Release FE + peer-window asymmetry	-8.151 (1.865)	-4.370	39.8%
Release FE + peer count	-2.107 (1.882)	-1.120	84.4%
Two-way fixed effects + both window terms	-0.003 (1.748)	-0.002	100.0%

Table 1: Placebo coefficient by conditioning set, in percentile points, always within release. Two-way provider-by-benchmark standard errors in parentheses (45 by 61 clusters; provider-only in Appendix B). $n = 2,831$ cells. Absorbed is the fall in the absolute coefficient from the first row; peer count alone is a suppressor, so the last row conditions on both window terms jointly.

The mechanism behind the second dimension is geometric. The comparison window is symmetric in calendar time, 182 days on each side of the release, but a benchmark’s model coverage is not symmetric in calendar time, because coverage begins when the benchmark is built. Figure 1 plots the two objects this creates against benchmark maturity at release: the share of the window that is newer falls with maturity and jumps at the availability boundary, while standing runs through the same boundary without a detectable break. The mechanism is discontinuous where the outcome is not, and peer-set composition is a property of timing alone.

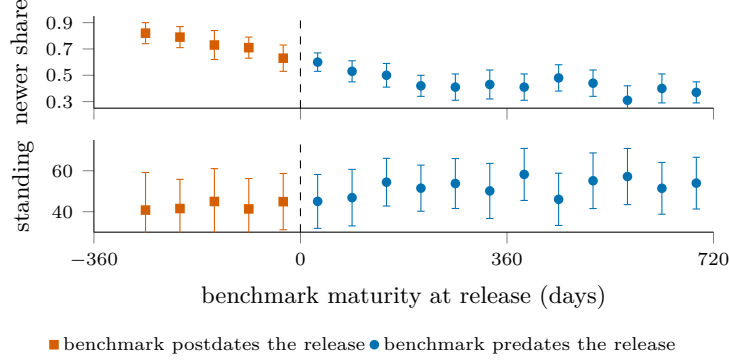


Figure 1: The mechanism and the outcome at the availability boundary. Top: the share of a cell’s comparison window released after the focal model, against benchmark maturity at release; it falls with maturity and jumps at the boundary. Bottom: within-benchmark standing on the same axis, continuous through it. Sixty-day bins; intervals clustered on organisation.

223 The share of a cell’s window released after the focal model is 0.4683 for eligible cells and
 224 0.7065 for placebo cells, against 0.5 for a two-sided window. Within release, the slope of
 225 standing on that share is -30.63 percentile points per unit among eligible cells alone. Fig-
 226 ure 2 plots standing against that share for both groups at once. Eligible and placebo cells
 227 do not form two clouds at different heights; they trace a single declining curve, common
 228 only once benchmark composition is absorbed (Table 1), and differ in where along it they
 229 fall. On the variable that assigns them, benchmark maturity at release, they cannot overlap
 230 at all, so the contrast is an extrapolation across that boundary rather than a comparison
 231 within common support. The check is at the boundary, the one place Section 2.1 leaves
 232 identified under continuity [Hahn et al., 2001]. Nothing detectable happens there. Local
 233 linear estimates are negative across four bandwidths and two polynomial orders, every inter-
 234 val covers zero though the widest reaches $+8.05$, and a window chosen on covariate balance
 235 puts the difference at -2.52 with a randomization p of 0.447 [Cattaneo et al., 2015]. The
 236 two groups’ mean maturity differs by 793 days, and the gap lives in that distance rather
 237 than at the boundary (grid and balance tests in Appendix B).

238 Standing decomposes into a weighted average of position among older and newer peers,
 239 percentile $= (1 - w) P_{\text{old}} + w P_{\text{new}}$ with w the newer share of window peers, exact cell by cell
 240 at double precision. Replacing the empirical weight with one half defines the side-balanced
 241 measure, available for 92.5% of all cells and 92.8% of eligible-or-placebo cells, and cuts the
 242 same slope to -8.56 .

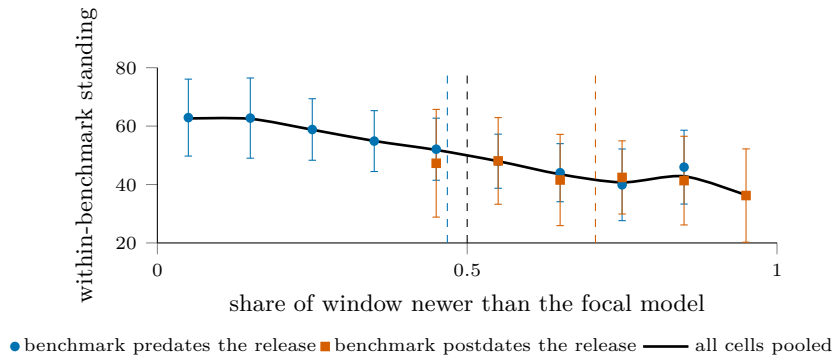


Figure 2: Standing against the share of a cell’s comparison window released after the focal model; dashed verticals mark each group’s mean share, and the black dashed line the symmetric 0.5. Eligible and placebo cells lie on one declining curve, so the contrast between them is a position on that curve rather than a fact about disclosure. Intervals are clustered on organisation.

That the identity holds does not make the gradient an accounting tautology. Shuffling scores within benchmark destroys the capability trend while leaving dates, window membership, peer counts and newer shares bit-identical. Under that null, the slope has mean -0.09 and standard deviation 3.12 . The observed -30.63 sits 9.8 standard deviations away, and none of 300 draws reaches it. Replacing scores instead with a within-benchmark linear fit on date plus permuted residuals, so that a capability trend exists and nothing else does, recovers 93% of the observed slope, averaged over 200 draws. Comparison-set geometry is inert on its own and becomes a bias only when it multiplies a real trend in capability.

3.3 Trimming and reweighting

The phenomenon has a name in nonparametric estimation: local averages are biased near the edge of a support, and the standard remedies recentre rather than discard [Gasser and Müller, 1979, Fan, 1992]. What is unusual is the boundary’s coincidence with the endowment: a benchmark was available to a release exactly when the model did not sit at the left edge of that benchmark’s coverage, so the indicator of what a provider could have held is the indicator of where the window is one-sided, and removing the edge removes the contrast. The coincidence has an older name, the age-period-cohort identity: release date, benchmark vintage and their difference are exactly collinear, and each has a measured signature here: t_i in the capability trend, τ_b in saturation, $t_i - \tau_b$ in window composition. Only the linear components are unidentified in that structure [Fosse and Winship, 2019a,b], so what fails is a slope. Three sign restrictions bound it: capability rises with the calendar, later benchmarks are less saturated, and standing rises with maturity. All three are assumptions on unidentified components, each motivated by a measurement of an identified quantity. Under them the unidentified component accounts for at most 24 to 34% of the placebo gap across specifications, roughly three percentile points, which is where the saturated regression also lands. The remainder is carried by the identified reduced-form slopes, and Appendix B gives the derivation.

Standing measure	Mean	Median	Positive	Sign-flip p	Releases	Cells
Windowed percentile, $\pm 182d$	+12.23	+10.69	78.1%	0.0001	146	100.0%
Rank against all models ever	+22.75	+22.66	89.0%	0.0001	146	100.0%
Trim to two-sided windows	+9.96	+10.69	72.2%	0.0001	115	70.4%
Side-balanced percentile	+9.14	+9.62	70.9%	0.0001	141	92.8%

Table 2: The label-free placebo null under each candidate measure: release-level mean standing of eligible minus postdating benchmarks, in percentile points. Every entry should be zero. Sign-flip p is the provider-clustered test of Section 2.5, against zero; Cells is the share of comparable cells where the measure is defined.

Table 2 shows what follows. Ranking against every model ever scored nearly doubles the contamination; trimming to two-sided windows discards 30% of the cells to remove 19% , and its row is just the uncorrected gap on the retained cells, composition rather than correction. Shortening the window does reduce the gap, to $+8.59$ at 91 days, but in proportion to the asymmetry it removes rather than by breaking the mechanism: across four widths and four benchmark-count minima the gap is positive in all 16 specifications, from $+8.59$ to $+21.66$, and tracks the eligible-minus-placebo gap in window composition at $r = 0.991$. Dropping any one of the 18 organisations leaves it between $+11.45$ and $+13.20$. The correction is a covariate, not a subsample.

One reading of that number would be too generous to it. Window composition is a function of the same dates that define the placebo indicator, so conditioning on it does not adjust for a confounder sitting outside the comparison; it splits the gap into the channel carrying it. A study that holds disclosure labels and wants an estimate of θ must first argue that window composition is settled before a provider decides what to report, or it will condition away part of what it is measuring. Provider identity explains little of the variance in window composition, which is suggestive rather than conclusive; Appendix B reports the decomposition and the coverage tilt it does not correct.

286 The side-balanced percentile is the one change of measure that does substantial work, cutting
 287 the placebo mean to +9.14 and the eligible-cell gradient to -8.56 . Figure 4 in Appendix B
 288 puts it against the uncorrected measure on the same cells, close to flat where cells are dense.
 289 Reweighting cannot reach what remains, since balancing repairs the weight placed on each
 290 side of a model but not which models an evaluator chose to score early, so evaluator attention
 291 stays a residual confound. It is undefined wherever a side is empty, so it accompanies the
 292 uncorrected measure rather than replacing it.

293 4 Evidence from HELM

294 Everything so far concerns a percentile constructed here on a panel assembled here, so the
 295 effect may belong to the construction rather than to leaderboards in general. HELM Lite
 296 publishes its full accuracy table at every versioned release, archives the earlier ones, and
 297 reports a pool-relative headline [Liang et al., 2023], so the same evidence can be read at 14
 298 points in time while the pool grows underneath it.

299 If a leaderboard moves, the first thing to rule out is a revised suite. Across releases v1.0.0
 300 through v1.13.0, the ten scenario columns are identical, with nothing added, dropped or
 301 renamed beyond one relabelled column header. The same 24 models appear in all 14 releases
 302 with their 240 published scenario scores bit-identical throughout, while the evaluated pool
 303 grows from 30 models to 91.

304 Against that frozen evidence, all 24 constant models see their published Mean win rate
 305 change, by 0.1232 on average in absolute value and by as much as 0.2178 for Mixtral 8x7B.
 306 Most of that movement is arithmetic: Mean win rate is the fraction of head-to-head compar-
 307 isons a model wins, so a fixed model faces a larger, later field as the pool grows; the change
 308 correlates with initial standing at -0.766 .

309 The movement that matters is elsewhere. If the drift were a monotone transform of the
 310 frozen evidence, it would be harmless, since a monotone rescaling preserves every order. It
 311 is not. Of the 24, 22 fall and two rise, and five of the 276 pairs among them, 1.8%, reverse
 312 between the first release and the last, each a published order inversion with bit-identical
 313 evidence beneath it. Figure 3 plots each model’s rank among the constant set at the first
 314 release against the last, so a reversed pair is a crossing, countable on the page while the
 315 evidence beneath is fixed. Unlike test-set reuse [Roelofs et al., 2019], the evidence is fixed;
 316 the reference set is what moves. The published numbers are correct as defined at every
 317 release. What changed was the pool, and the published order followed the pool. Singh
 318 et al. [2025] reports a comparable pattern from a different mechanism, deprecation policy
 319 rather than a coverage boundary changing which models remain available, and in both the
 320 reference set rather than the evidence moves the ranking.

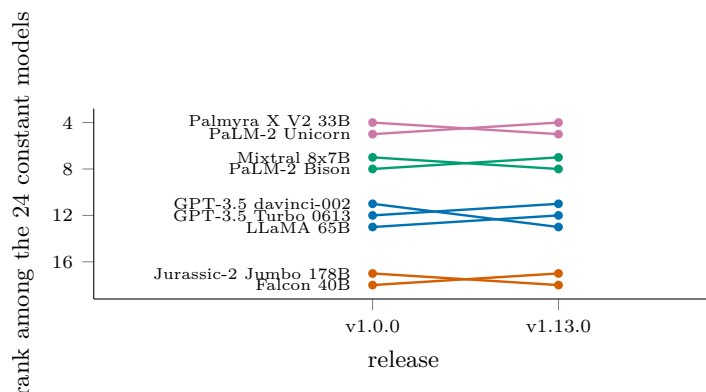


Figure 3: The nine HELM Lite models whose published order changed between the first release and the last, ranked among the 24 whose 240 scenario scores are bit-identical throughout; coloured by group of mutually reversing models, four groups and five reversed pairs in all. The control: across 16 HELM Capabilities releases behind an absolute headline, with the pool growing from 22 to 68, none of the 22 constant models’ headline values moved and no pair reversed.

The control, HELM Capabilities, comes from the same organisation, whose maintainers had diagnosed the win rate as pool-dependent [Xu et al., 2025]; it runs on the same infrastructure and cadence, its pool growing comparably, from 22 models to 68 over 16 releases. Its headline is an absolute mean over scenarios rather than a fraction of comparisons won. The same 22 models appear in every release with 110 scenario scores bit-identical, none of the 22 headline values moves, and no pair reorders, at the endpoints or ever. An absolute mean over bit-identical scores on a fixed suite cannot move with the pool, so zero drift is arithmetic, a scope condition rather than a test that could have failed.

This need not be read as an error on HELM’s part: the absolute headline on its own Capabilities leaderboard is the point. A pool-relative headline is a joint statement about a model and about the population evaluated beside it; read later as statements about the models alone, two such numbers move standing between providers for reasons belonging to neither. Restoring comparability across moments is the classical equating problem [Kolen and Brennan, 2014], which an absolute scale estimated jointly across benchmarks avoids entirely [Ho et al., 2025]. The panel and the leaderboard fail in the same way: the reference set rather than the evidence moves the number.

5 Coding the Record

A benchmark can be missing from a model card because it was irrelevant, never run, or run and withheld. Only the third is disclosure in the sense of Grossman [1981], Milgrom [1981] and Dye [1985], and only it is a reader’s to price. The within-family drop narrows it: a provider that reports b at t and omits it at $t + 1$, an independent score present, has established relevance and possession of the harness by its own earlier behaviour, with the independent score as the registry-style counterfactual [Chan et al., 2004, Dwan et al., 2013, Kirkham et al., 2010]. It does not establish that b was run for $t + 1$, so the Dye [1985] uncertainty narrows rather than closes. Its ceiling, 37 pairs, and detection threshold, roughly 38 points at one drop, sit in Appendix B.

A single coder read the release-time document of each of the 100 releases for which one is locatable; eight of the queued 108 have none. The record notes every benchmark a document reports, along with variants, bare mentions, and stated benign absences, and the ORBIT categories are derived mechanically from these records and the family history. This yields 1,280 coded cells, of which 31 belong to five later-vintage releases outside the pinned panel and are excluded. An independent second coder then recoded a provider-stratified fifth of the releases, blind to the first coding and without calibration contact; Appendix B records the one release where that blindness broke. Agreement is Cohen’s κ over the derived cells [Cohen, 1960], with intervals from resampling releases. The two codings agree at 0.95 [0.80, 1.00] on the primitive judgment of whether a benchmark was reported, at 0.85 [0.72, 0.95] over the full category set, and at 0.81 [0.48, 1.00] on the high-versus-low-suspicion split. The single E cell each coding derives is the same cell, and Appendix B reports the confusion structure. Among the 1,249 eligible coded cells that resolve in the panel, 76.9% report no number, and the release-level mean is 73.1% [65.9, 79.2]. The rate is descriptive and is not, by itself, evidence of selection.

Standing measure	Coded gap	95% CI	No labels	Difference
Windowed percentile, $\pm 182d$	+11.40	[+5.6, +17.0]	+14.13	−2.73
Side-balanced percentile, $\pm 182d$	+7.90	[−0.4, +15.2]	+9.82	−1.92
Rank against all models ever	+22.33	[+18.0, +28.1]	+24.50	−2.17
Windowed percentile, $\pm 90d$	+8.15	[+1.3, +14.7]	+10.36	−2.21
Windowed percentile, $\pm 365d$	+18.03	[+13.9, +23.1]	+20.53	−2.50

Table 3: The coded omission deficit beside the label-free gap, in percentile points, over 71 releases and 13 providers (70 side-balanced). No labels is the same contrast with no disclosure coding, on the same releases; Difference is Coded gap minus No labels. Intervals are provider-clustered bootstraps; four of five coded gaps exclude zero.

Table 3 is the section’s result: the disclosed-versus-omitted comparison returns +8.7 points [4.1, 13.9] over 63 releases; its identifying counterpart, omitted-eligible against postdating

cells over the 71 releases holding both, returns +11.4 [5.6, 17.0], significant against zero at $p = 0.003$. Read against zero it would be concealment; read against Section 3 it is the availability gap: computed on the same releases with no labels, the contrast sits at +14.1, and under every measure and window the coded gap falls 1.9 to 2.7 points short of its matched counterpart. The shortfall is a point-estimate decomposition, the 23% reported share times the disclosed-omitted gap, so the labels change this contrast only through composition; intervals spanning eleven points leave a strategic component of several points undetected rather than excluded. The naive reading is robust to two-way provider-and-benchmark errors, to reclassifying every variant cell as an omission, and to any leave-one-out, because the availability gap is. What survives instead is composition. Reported benchmarks stand 8.7 points above omitted eligible ones within release, a composition gap larger than all 9,999 count-preserving random assignments ($p = 0.0001$, the floor the draws allow; null standard deviation 1.9) and one the availability gap cannot produce. Benchmark identity does not carry it: with benchmark fixed effects beside the release effects the coefficient rises from +7.4 to +9.3, and conditioning on the window share leaves +7.4. It does not identify whether standing drove the reporting: relevance, salience, and within-eligible contamination can move disclosure and performance together. The drop estimator realises 16 drops across 15 releases and returns +2.9, sign-flip $p = 0.25$ over five providers, the underpowered null its calibration priced. Appendix B adds three unrepaired limits: unlocatable documents, coverage that may follow provider emphasis, and the best-across-scaffolds rule.

6 Conclusion

We set out to measure how much of what providers could report they choose not to. The coded record returns the number and the reason to distrust it: +11.4 points, significant against zero, yet short of its matched label-free counterpart by only what composition implies, and too imprecise to exclude a strategic component of a few points. Dating the instruments bounds the endowment, though the bound is an assumption about access, and the same dates fix where a model sits inside a benchmark’s coverage: what identifies the endowment moves the measure used to price silence, and reweighting is a partial defence. Growing the drop record, 16 drops against 37 qualifying pairs, is a reporting-standard question, answerable in two ways: a jointly estimated scale [Ho et al., 2025], which shrinks the placebo gap without closing it, and carrying a benchmark set across a family’s releases, narrowing the empty-endowment excuse and the Dye [1985] pooling at no disclosure cost.

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499 A What a Provider Could Report

500 The set of results a provider could publish is not fixed, and it grows. When we count the
 501 benchmarks in our panel by the year each was published, the dated stock rises from 11 at
 502 the end of 2019 to 62 by 2026. A benchmark that enters that stock stays available to every
 503 release after it, so the denominator of any omission rate is weakly increasing by construction.

504 What a single release could report grows more slowly and less evenly. Averaged by half-year,
 505 the number of eligible benchmarks per release runs 6.8 in the first half of 2023, falls to 3.9
 506 in the first half of 2024, a period in which releases outpaced the instruments now used to
 507 score them, and reaches 10.1 by the first half of 2026. The two series move together over
 508 the sample and apart within it, which is consistent with the opportunity set being shaped
 509 partly by release timing rather than by provider choice.

510 A count of omissions therefore measures the calendar. Unless reported tables expand one
 511 for one with the stock, omission rates drift upward for arithmetic reasons alone, and com-
 512 paring early releases with late ones on that measure compares eras rather than providers.
 513 The question that survives is composition: which benchmarks are dropped, conditional on
 514 availability, against what an indifferent selection over the same set would have produced.
 515 Answering it requires knowing not only what was available but how the release stood on
 516 each available benchmark, and it is that second requirement, not the first, that the rest of
 517 the paper shows to be the hard one [Dranove and Jin, 2010].

518 B Additional Checks

519 Is window composition a provider’s choice? Section 3.3 conditions on window composition
 520 and notes that doing so is defensible only if composition is settled before a provider decides
 521 what to report. Regressing the share of a cell’s window released after the focal model on
 522 provider dummies gives an R^2 of 0.071 across 45 organisations, against 0.002 for benchmark
 523 dummies over 61 benchmarks and 0.104 for the release date alone. Window composition is
 524 close to idiosyncratic at the cell level, which is consistent with its being set by the evaluator’s
 525 scoring schedule rather than by the provider, and is not enough to establish it.

526 Is the evaluator’s coverage selective? The estimand conditions on the cells the evaluator
 527 scores, so coverage is upstream of everything else. Coverage averages 8.7 benchmarks per
 528 release with a standard deviation of 8.0 and a range of 1 to 38. It correlates with a provider’s
 529 release count at 0.203 and with the release date at 0.265, and releases from the densest
 530 quartile of providers carry 9.6 scored benchmarks against 8.4 for the rest. Coverage is
 531 therefore mildly tilted toward larger and later providers, and nothing in this paper corrects
 532 for that.

533 Bounding the unidentified component. Write t for a release date, τ for a benchmark’s
 534 vintage and $a = t - \tau$ for the benchmark’s maturity when the model met it. A regression
 535 of standing on all three is singular by construction, which is the age-period-cohort problem,
 536 and the data identify only $\Pi_p = \beta_a + \beta_t$ and $\Pi_c = \beta_\tau - \beta_a$, leaving a one-parameter family
 537 indexed by $\lambda = \beta_a$. Fitting the reduced form on the 2,831 comparable cells gives $\Pi_p = +4.12$
 538 and $\Pi_c = -1.36$ percentile points per year.

539 Three restrictions come from the paper rather than from outside it. Capability rises over
 540 calendar time, so $\beta_t \geq 0$. Later benchmarks are less saturated, which Section 3.2 measures
 541 directly, so $\beta_\tau \leq 0$. Standing rises with maturity, since a model on a young benchmark is
 542 ranked against newer peers, so $\beta_a \geq 0$. Together these give $0 \leq \lambda \leq \min(\Pi_p, -\Pi_c) = 1.36$,
 543 and saturation rather than capability is the binding restriction. The mean maturity gap
 544 between eligible and placebo cells is 2.17 years, so the unidentified component can account
 545 for at most $1.36 \times 2.17 = 2.94$ percentile points, 24% of the observed gap. Adding centred
 546 quadratic and cubic terms in t and τ , which are not collinear with a , moves the ceiling to
 547 34% and 31%.

548 Shuffling scores within benchmark leaves every date and the rank deficiency untouched while
 549 destroying the capability trend. Under that null the same quantity falls to 0.02 percentile

550 points on average over 30 draws, with a maximum of 0.24, so the bound reflects the trend
 551 the geometry multiplies rather than the geometry alone.

552 The boundary as a discontinuity. Eligibility is $1[t_i > \tau_b]$, a deterministic function of the
 553 two dates, and that is what the global repairs run into. Equating across groups that took
 554 different instruments assumes a conditional relationship common to both, which two groups
 555 defined by the boundary cannot have [Sinharay and Holland, 2010], and reweighting by
 556 the probability a cell is observed needs that probability bounded away from zero, which a
 557 deterministic rule denies [Ma and Chen, 2019]. The same determinism makes the boundary a
 558 sharp discontinuity, where standing at the threshold is identified under continuity whatever
 559 happens globally [Hahn et al., 2001]. Local linear estimates with a triangular kernel and
 560 provider-clustered errors give -8.82 (4.76) at ± 60 days, -6.85 (4.09) at ± 90 , -4.08 (3.11)
 561 at ± 180 and -2.85 (2.35) at ± 250 , and a quadratic fit at each width moves none of them
 562 across zero. All eight are negative, all eight intervals cover zero, and the largest upper bound
 563 is $+8.05$, below the $+12.23$ the global comparison reports.

564 The local randomization framework agrees [Cattaneo et al., 2015]. Widening symmetric
 565 windows and testing the release’s benchmark count, scaffold count and scaffold spread for
 566 balance, we find the smallest of the three p -values to be 0.335 at ± 20 days and 0.571
 567 at ± 40 before falling to 0.058 at ± 60 , which selects ± 40 , a rule applied after seeing the
 568 balance results, so the randomization p below does not adjust for the selection. The same
 569 difference is -5.96 ($p = 0.23$) at ± 20 days and $+0.43$ ($p = 0.88$) at ± 60 . That window holds
 570 113 postdating and 114 eligible cells, and the difference in mean standing is -2.52 with a
 571 randomization p of 0.447. Cell counts in the 30 days either side of the cutoff stand at 80
 572 against 84, a ratio of 0.95, so the running variable is not bunching against the threshold
 573 [McCrary, 2008].

574 The mechanism is discontinuous where standing is not. The share of a cell’s window released
 575 after the focal model jumps $+0.083$ at the cutoff, which at the eligible-cell gradient of -30.63
 576 implies about -2.5 percentile points, inside every interval above, so at the cutoff the design
 577 separates concealment from the availability gap only above that effect, and the narrowest
 578 windows cannot reject the full -12.23 either. Peer count and raw score are continuous.
 579 The global gap therefore lives in the 793 days that separate the two groups’ mean maturity
 580 rather than in anything happening at the boundary itself.

581 A standing measure with no comparison window. The conclusion points at a scale esti-
 582 mated jointly across benchmarks rather than against contemporaries, which removes the
 583 comparison window from the measure entirely. Fitting one does not remove the gap. Abil-
 584 ity and difficulty are estimated together over the 63 benchmarks reported on a common
 585 zero-to-one scale, on a logit link, with the loading ridged toward one because a free load-
 586 ing is unidentified at this density and takes slopes past 65,000. Restricting to benchmarks
 587 carrying at least twenty releases and releases carrying at least three benchmarks leaves
 588 1,669 eligible cells over 31 benchmarks. The model is fit on eligible cells alone, so the 226
 589 postdating cells it can score are out of sample.

590 The eligible-minus-placebo gap in the residuals is $+0.336$ residual standard deviations,
 591 against $+0.449$ for the windowed percentile on the same contrast, positive in 64.0% of
 592 the 89 releases holding both kinds, at $p = 0.005$. Removing the window shrinks the gap by
 593 about a quarter and leaves it positive, which puts a jointly estimated scale beside trimming
 594 at 19% and side-balancing at 25%, each on its own scale, rather than ahead of them. Its
 595 difficulty parameters are estimated from the maturity-selected cells that exist, so part of the
 596 surviving gap may be the same selection re-expressed, and its placebo cells are scored out of
 597 sample while every eligible cell is in sample, an asymmetry the contrast inherits. Shuffling
 598 scores within benchmark, which leaves every date and the rank deficiency untouched, drops
 599 the same quantity to 0.02 on average over 30 draws.

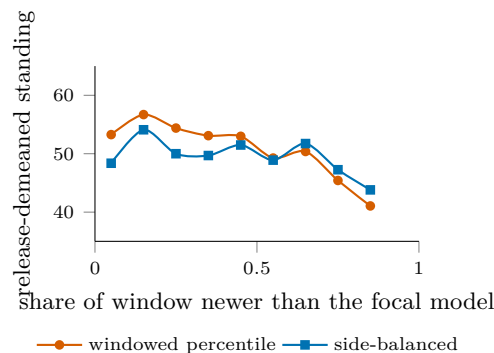


Figure 4: The side-balanced percentile against the windowed one, on eligible cells within release. The windowed measure slopes across the range while the side-balanced one is close to flat where cells are dense; the quoted gradients, -30.63 to -8.56 per unit share, are the within-release regression slopes on these cells, and the release-demeaned binned means shown are correspondingly flatter.

600 A backward-only standing, position among older peers alone, removes the window's forward
601 half entirely and leaves the label-free contrast at $+12.11$ across 142 releases, so the gap does
602 not come from look-ahead.

603 Drop-design ceiling and calibration. The panel contains 144 adjacent same-family release
604 pairs covering 711 shared eligible benchmarks and 15 organisations. Requiring both releases
605 in a pair to carry at least eight scored benchmarks leaves 37 pairs, 482 shared benchmarks,
606 and 8 organisations, the effective cluster count, in the regime where Section 2.5 reports no
607 analytic standard error. Under the null of no selective withholding, the sampling standard
608 deviation of a drop gap, over 400 draws for each of 117 qualifying releases, is 19.4 percentile
609 points when one benchmark is dropped, 14.4 when two are and 12.4 when three are, so at
610 conventional two-sided size a single-drop event must move the gap by roughly 38 points
611 before it separates from noise, three times the contamination itself, and no single drop can
612 be read alone.

613 Reliability detail. The second coding covers 20 of the 100 readable releases, drawn strat-
614 ified by provider. Derived categories import contemporaries' evidence across releases, a
615 dependence the release resampling does not model. Across 15 of them, 194 derived cells
616 pair: one drawn release is a zero-disclosure document whose blank evidence rows the pair-
617 ing drops, a conservative exclusion since all eleven of its cells are trivial agreements, and four
618 drawn releases sit outside the pinned worklist. The reported-versus-not cross-tabulation is
619 one-directional, 47 jointly reported and 143 jointly not: no cell the first coder called reported
620 was coded otherwise by the second, and the four reversals all sit on one release whose second-
621 coding note records retaining four contested scores despite the first coder's disagreement,
622 an exposure that breaks blindness for that release. Excluding it, reported-versus-not agree-
623 ment is exact and the full-category κ is 0.88 on 182 cells. Of the 14 disagreements overall,
624 seven are the variant boundary between C and A, one a bare mention against a value, two
625 the derived split between G and H on a single release, and four the exposed release above.
626 Three limits of the record are candidly unrepaired: the eight releases without a locatable
627 document are unlikely to be missing at random and plausibly document less; the evaluator's
628 coverage may itself respond to what providers emphasise, which would tilt the reported set
629 upward by sampling; and the best-across-scaffolds score rule is not probed for asymmetry
630 across the boundary. High-suspicion marginals match exactly at 5.7% for each coder, which
631 is why the split κ holds at 0.81 despite the category's rarity; its wide interval reflects eleven
632 such cells concentrating in a few releases.

633 Provider-only clustering, for comparison. The manuscript's regressions cluster on provider
634 and benchmark jointly. Provider-only errors are consistently smaller: 1.271 against 2.038
635 on the release-effects row, a ratio of 1.60, and 1.652 against 1.748 on the fully saturated
636 row, a ratio of 1.06. The benchmark dimension carries real dependence on the raw contrast
637 and almost none once benchmark effects and the window terms are in, which is what the

638 decomposition says it should. The headline sign-flip test at $p = 0.0001$ preserves provider
639 dependence only; the two-way regression row above is the statement that the contamination
640 survives both dimensions, at $t = -6.6$.

641 NeurIPS Paper Checklist

642 1. Claims

643 Question: Do the main claims made in the abstract and introduction accurately
644 reflect the paper’s contributions and scope?

645 Answer: [\[Yes\]](#)

646 Justification: The abstract and introduction state both findings within their identi-
647 fied scope: the coded omission gap is indistinguishable from the label-free gap, and
648 within-release composition tracks standing. Section 5 reports the reliability of the
649 coding behind them.

650 2. Limitations

651 Question: Does the paper discuss the limitations of the work performed by the
652 authors?

653 Answer: [\[Yes\]](#)

654 Justification: Limitations appear where they bear on the argument rather than
655 only at the end. Section 2.1 bounds the estimand and states that the reported
656 omission rate is descriptive and identifies no withholding; Section 3.2 states that
657 the conditioning is a decomposition rather than a control strategy, and what a
658 study holding disclosure labels would additionally have to argue; Section 3.3 reports
659 the corrections that fail and the coverage the balanced measure loses; Section 5
660 reports the ceiling on the design that narrows endowment uncertainty furthest, its
661 16 realised drops, and the underpowered null they return.

662 3. Theory assumptions and proofs

663 Question: For each theoretical result, does the paper provide the full set of assump-
664 tions and a complete (and correct) proof?

665 Answer: [\[Yes\]](#)

666 Justification: Section 2.1 states one identification result, that a temporal eligibility
667 gate cannot satisfy the exclusion restriction the design needs, and gives its argument
668 in full: the three date variables are exactly collinear, so excluding one requires
669 that at least two of them not move standing. Section 3 shows all three move
670 it. Appendix B states the three sign restrictions used to bound the unidentified
671 component and derives the bound from them. Everything else in Section 2.1 restates
672 published results with citation.

673 4. Experimental result reproducibility

674 Question: Does the paper fully disclose all the information needed to reproduce
675 the main experimental results of the paper to the extent that it affects the main
676 claims and/or conclusions of the paper (regardless of whether the code and data
677 are provided or not)?

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679 Justification: Every number in the paper is written by one command into a single
680 machine-readable file, and the data snapshot it was computed against is pinned
681 by a per-file SHA-256 manifest in the released code. Section 2 states the release
682 key, the eligibility rule, the window width, and which estimates impose a minimum
683 benchmark count and which do not.

684 5. Open access to data and code

685 Question: Does the paper provide open access to the data and code, with sufficient
686 instructions to faithfully reproduce the main experimental results, as described in
687 supplemental material?

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689 Justification: Code and derived data are released with instructions. The indepen-
690 dent scores are redistributed by their publisher under CC-BY and are fetched rather
691 than copied, then verified against the pinned manifest, so an upstream change is
692 reported rather than silently absorbed. The external leaderboard evidence is frozen

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6. Experimental setting/details

Question: Does the paper specify all the training and test details (e.g., data splits, hyperparameters, how they were chosen, type of optimizer) necessary to understand the results?

Answer: [Yes]

Justification: There is no model training. The choices that affect the estimates are stated in Section 2: the release key, the rule for collapsing scaffold variants, the 182-day comparison window, which estimates impose a minimum benchmark count and which do not, and the treatment of pairs whose benchmark vintage is unverified.

7. Experiment statistical significance

Question: Does the paper report error bars suitably and correctly defined or other appropriate information about the statistical significance of the experiments?

Answer: [Yes]

Justification: Section 2.5 states the inference protocol. The cluster count is small and unevenly distributed, so below twelve clusters no analytic standard error is reported, coefficients in that range use a restricted wild cluster bootstrap, release-level means use a cluster sign-flip randomization test, and no result is called significant on a cluster-t alone. Table 1 reports a standard error with every coefficient, the headline contrast carries a randomization p-value, and the permutation null is reported with its mean, its spread and the number of draws reaching the observed value.

8. Experiments compute resources

Question: For each experiment, does the paper provide sufficient information on the computer resources (type of compute workers, memory, time of execution) needed to reproduce the experiments?

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Justification: All analysis runs in seconds on a laptop. There is no training, no accelerator use, and no discarded run.

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Question: Does the paper discuss both potential positive societal impacts and negative societal impacts of the work performed?

Answer: [Yes]

Justification: Section 4 states the consequence for a reader of a pool-relative leaderboard, namely that such a number describes standing within one vintage of the evaluated pool rather than a property of the model. The conclusion states that whether the record needed to measure withholding should grow is a reporting-standard question. The paper does not attribute concealment to any named provider, because the design does not support that.

11. Safeguards

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748 Question: Are the creators or original owners of assets (e.g., code, data, models),
749 used in the paper, properly credited and are the license and terms of use explicitly
750 mentioned and properly respected?
751 Answer: [Yes]
752 Justification: The independent scores are credited to their publisher and used under
753 CC-BY, and the external leaderboard is cited to its published description. Both are
754 named in the text and in the released code.

755 13. New assets
756 Question: Are new assets introduced in the paper well documented and is the
757 documentation provided alongside the assets?
758 Answer: [Yes]
759 Justification: The released code documents how each number is produced, the snap-
760 shot manifest records the exact data build, and the frozen external evidence records
761 a source URL, a retrieval date and a SHA-256 for every file.

762 14. Crowdsourcing and research with human subjects
763 Question: For crowdsourcing experiments and research with human subjects, does
764 the paper include the full text of instructions given to participants and screenshots,
765 if applicable, as well as details about compensation (if any)?
766 Answer: [N/A]
767 Justification: The work involves no crowdsourcing and no human subjects.

768 15. Institutional review board (IRB) approvals or equivalent for research with human
769 subjects
770 Question: Does the paper describe potential risks incurred by study participants,
771 whether such risks were disclosed to the subjects, and whether Institutional Review
772 Board (IRB) approvals (or an equivalent approval/review based on the requirements
773 of your country or institution) were obtained?
774 Answer: [N/A]
775 Justification: The work involves no human subjects, so no review board approval
776 was required.

777 16. Declaration of LLM usage
778 Question: Does the paper describe the usage of LLMs if it is an important, original,
779 or non-standard component of the core methods in this research? Note that if the
780 LLM is used only for writing, editing, or formatting purposes and does not impact
781 the core methodology, scientific rigor, or originality of the research, declaration is
782 not required.
783 Answer: [Yes]
784 Justification: An earlier pass used a large language model to assign the disclosure-
785 coding categories; those labels were cleared and none enters. The categories now
786 standing were hand-assigned by a human coder from archived release-time docu-
787 ments under the frozen protocol, with an independent second coder blind-coding
788 a provider-stratified 20% sample (Section 5), so no number reported here depends
789 on a model’s judgment. Every number is produced by the released code from the
790 pinned snapshot. Language models were otherwise used only for writing and code
791 assistance.